REQUIREMENT ANALYSIS
Solution Requirements

Date	01 JULY 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Solution Requirements

The Solution Requirements define the functional and non-functional capabilities required for the successful implementation of the Electricity Consumption Analysis project. These requirements ensure that the system meets user expectations, supports interactive data analysis, and delivers reliable insights through dashboards and web-based visualization.

Functional Requirements

Following are the functional requirements of the proposed solution.

FR No	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Dashboard Visualization	Display an interactive Tableau dashboard with electricity consumption KPIs such as Total Usage, Average Usage, Highest Usage, and Lowest Usage.
FR-2	State-wise Electricity Analysis	Analyze electricity consumption across different states and compare state-wise usage patterns.
FR-3	Region-wise Electricity Analysis	Compare electricity usage among Northern, Southern, Eastern, Western, and Central regions.
FR-4	Monthly Trend Analysis	Display monthly electricity consumption trends and identify seasonal demand variations.
FR-5	Geographic Map Analysis	Display electricity consumption on an India map using state locations for better geographic understanding.
FR-6	Top 10 Electricity Consumers	Identify and display the top ten highest electricity-consuming states using interactive charts.
FR-7	Interactive Filters	Allow users to filter dashboard data by State, Region, and Date for customized analysis.
FR-8	Tableau Story	Present a guided Tableau Story explaining electricity consumption trends, regional comparisons, and business insights.

FR No	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-9	Flask Web Application	Embed the Tableau Dashboard and Tableau Story into a Flask web application for easy browser access.
FR-10	Business Insights	Generate summarized insights and conclusions to support electricity boards and government decision-makers in energy planning.

Non-Functional Requirements

Following are the non-functional requirements of the proposed solution.

NFR No	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard should be simple to use with clear navigation, readable charts, and interactive filters for all users.
NFR-2	Security	The Flask application should securely present electricity analysis without exposing or modifying the original dataset.
NFR-3	Reliability	The dashboard should consistently display accurate calculations and visualizations based on the electricity consumption dataset.
NFR-4	Performance	Dashboard pages, filters, and visualizations should load quickly and respond smoothly to user interactions.
NFR-5	Availability	The Tableau Public dashboard and Flask web application should be available whenever users need to access electricity analysis.
NFR-6	Scalability	The system should support larger electricity datasets, additional dashboards, and future analytical features without major redesign.

Conclusion

The Solution Requirements provide a clear specification of the functional and non-functional capabilities required for the **Electricity Consumption Analysis** project. These requirements ensure that the system delivers accurate analysis, interactive visualizations, secure access, high performance, and scalability. By fulfilling these requirements, the project provides an efficient platform for analyzing electricity consumption patterns and supports informed decision-making for energy management and planning.